Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Senior Full Stack Developer with 8 years of professional experience building complex, production-grade web applications in TypeScript across healthcare, fintech, civic tech, and construction tech environments. Deep expertise in Node.js backend development and React frontend architecture, with hands-on experience designing AI-powered features including RAG systems, multi-agent pipelines, and conversational interfaces. Proven track record of end-to-end feature ownership, monorepo architecture (Nx, pnpm workspaces), CI/CD automation, and comprehensive E2E testing (Playwright, Cypress) in agile team environments.

CORE COMPETENCIES

- **Full Stack Development:** TypeScript, Node.js, NestJS, React, Next.js, REST APIs, GraphQL, JWT authentication, and RBAC.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, and agentic workflows.
- **Monorepo & Architecture:** Nx, pnpm workspaces, modular package design, SSR/SSG, design systems, and scalable component systems.
- **Databases & Data:** PostgreSQL (pgvector), MongoDB, Prisma, real-time streaming (SSE, EventSource), and data visualization (Recharts, react-chartjs).
- **Testing & Quality:** Playwright, Cypress, Jest, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, GitHub Actions, CI/CD pipelines, Vercel, and Heroku.
- **Performance & Observability:** Core Web Vitals, code splitting, CDN strategies, Lighthouse, Sentry, Langfuse, and bundle analysis.
- **Leadership:** End-to-end feature ownership, architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension** in **TypeScript**, owning the feature end-to-end from architecture through deployment.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational AI interface.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, securely handling tokens with HTTP-only cookies and collaborating with backend engineers on auth strategies.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Introduced **Redux** for state management and **React Query** for server-state caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios**.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 TypeScript applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 interactive chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm workspaces**.
- Designed and implemented a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback), powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Built the **NestJS** backend with **PostgreSQL 16**, **Prisma v7**, and **pgvector** extension for vector storage (708K+ chunks, 1024-dim embeddings via Voyage AI), implementing **SSE streaming** for real-time chat responses with 15s keepalive heartbeats and token-budgeted context management.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** reverse proxy routing.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing 700+ documents across 37 states with zero ingestion errors. Integrated **Langfuse** for LLM observability and **Infisical** for secrets management.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*